

Meghana
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Senior Python Developer

PROFESSIONAL SUMMARY

- **Web Framework Expertise:** 8 years of experience with Python and Django, implementing robust web frameworks like Flask and FastAPI for efficient application development.
- **API Development:** Proficient in developing and deploying RESTful and GraphQL APIs, leveraging Django and FastAPI for high-performance, scalable web services.
- **Front-End Integration:** Skilled in creating interactive and responsive front-end solutions using HTML, CSS, JavaScript, Bootstrap, jQuery, and React, ensuring seamless integration with back-end services.
- **Database Management:** Extensive experience with relational databases (PostgreSQL, MySQL, SQLite) and NoSQL databases (MongoDB, Cassandra), optimizing data models and writing complex queries for high performance and data integrity.
- **Authentication and Security:** Expertise in implementing secure authentication mechanisms using JWT tokens and OAuth2, ensuring robust data privacy and user authentication.
- **Data Analysis and Machine Learning:** Proficient in using Python libraries like NumPy, pandas, Matplotlib, Keras, scikit-learn, and seaborn for data analysis, visualization, and machine learning tasks.
- **CI/CD and DevOps:** Experience in developing CI/CD pipelines using AWS Cloud Architecture, Docker, Jenkins, and Kubernetes, automating testing, deployment, and scaling of applications.
- **Cloud Deployment:** Skilled in deploying and managing web applications on AWS (EC2, S3, RDS, Lambda, Elastic Beanstalk), Azure, and GCP, ensuring scalable and reliable cloud-based solutions.
- **Version Control and Collaboration:** Extensive experience with Git for version control, facilitating team collaboration and efficient code management.
- **Testing and Quality Assurance:** Proficient in using PyTest and unit testing methodologies, ensuring robust and error-free applications through Test-driven Development (TDD) and Integration testing.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript
Python Framework	Django, Flask, Django Rest framework, FAST API
Web Technologies	HTML, CSS, JSON, AJAX, MVC, MVT, RESTful
JavaScript Libraries	React.js, jQuery
CSS framework	Bootstrap
Database	MySQL, Postgres, NoSQL
IDE's & Tools	Visual Studio, PyCharm, Jupyter Notebook, Google Colab.
Cloud Platforms	AWS, Microsoft Azure, GCP
Testing Tools	Pytest, Unit Test, Postman, Insomnia
Version Control	Git
Containerization & Orchestration	Docker, Kubernetes
CI/CD Tools	Jenkins
Deployment	Heroku, AWS, Azure
Package Managers	Pip, npm, yarn, Pipenv
Python libraries	Pandas, NumPy, matplotlib, cv2, scikit-learn, Keras, os, PyArrow

PROFESSIONAL EXPERIENCE

Sr, Python developer

Responsibilities:

- Developed backend modules using Python on Django Web Framework by implementing MVC architecture.
- Utilized AWS services such as Lambda, S3, and RDS to develop and deploy Python-based applications for efficient cloud-based data processing and storage.
- Implemented data extraction, transformation, and loading (ETL) processes using Python scripts and AWS Glue, ensuring seamless data integration from various sources into AWS Redshift.
- Developed REST APIs using Python 3.x and FastAPI, deployed using Docker containers managed by AWS ECS for scalable and reliable web services.
- Designed and implemented scalable ETL pipelines using PySpark for processing large volumes of structured and unstructured data from various sources like HDFS, S3, and Kafka.
- Integrated Quartz Scheduler for automating and managing scheduled tasks in Python applications, ensuring timely execution of background jobs.
- Leveraged AWS CodePipeline, CodeBuild, and CodeDeploy to establish a comprehensive CI/CD pipeline, automating testing, and deployment of Python applications.
- Utilized AWS CloudFormation and Terraform for infrastructure as code (IaC) to provision and manage AWS resources, ensuring consistency and scalability in deployment.
- Employed Amazon S3 for static and media file storage, ensuring secure and efficient data access and retrieval.
- Designed and developed highly scalable and distributed systems using AWS services, collaborating with architects to implement RESTful web services and FastAPI design, ensuring efficient and reliable system performance.
- Extensive experience with PySpark for distributed data processing, including DataFrame manipulation, RDD transformations, and actions.
- Implemented Quartz job persistence using database backends, enabling the recovery of scheduled tasks and triggers after application restarts.
- Developed tools using Python Django and utilized MongoDB for databases, writing parsers in Python for extracting useful data from the design database and performing ETL extraction.
- Extensively worked with Spark DataFrames on AWS EMR for ingesting data from flat files into RDDs to transform unstructured and structured data.
- Utilized React.js for creating interactive pages, employing one-way data flow, Virtual DOM, and JSX for front-end development.
- Established a comprehensive CI/CD pipeline by integrating SCM (Git) with Gradle for automated testing, deploying through Jenkins with Docker containers in production, and leveraging several DevOps tools such as AWS CodePipeline, Terraform, and Kubernetes.
- Automated data processing workflows using Apache Airflow and PySpark, ensuring efficient and reliable execution of data pipelines.
- Used JIRA for requirement analysis, task assignment, and to provide review comments.
- Developed reusable and customizable components for the new website, utilizing React.js for front-end development and React-Router for creating a single-page web application.
- Developed a Wrapper in Python for instantiating multi-threaded applications.
- Fixed bugs, provided production support, and enhanced applications by improving code reuse and performance using various design patterns.

SR. Python Developer**Citizens Bank, Phoenix, AZ****Aug 2021 - July 2023****Responsibilities:**

- Understand the user requirements and breakdown the tasks accordingly to implement in timely manner.
- Utilized Django Web Framework for creating web applications utilizing model view control engineering

- Build the REST API's using Django and integrate with the Angular app on AWS service.
- Utilized AWS Lambda for executing backend processes and event-driven functions, ensuring efficient and cost-effective compute resource management.
- Experienced working with Django and PostgreSQL.
- Worked with Django user management modules to create authenticated application
- Experienced working with GIT, version control to commit the changes and follow the workflow.
- Wrote Python scripts to parse JSON documents and load the data in PostgreSQL database.
- Worked on the backend application by using Django Web Framework to create API's.
- Integrated AWS Lambda functions with API Gateway to create and deploy serverless REST APIs, enhancing scalability and reducing operational overhead.
- Knowledge in integrating the backend and front, sorting the issues and bugs during integration.
- Worked with JIRA to manage the issues, track the bugs and prioritize the issues.
- Used Django API's to access the database
- Experience in working with TDD/BDD environments and Scrum development lifecycle.
- Worked with IDE's and also testing the API calls using Postman before integrating with the front end.
- Experienced in documenting the API's using tools like swagger.
- Good knowledge of scrum and agile methodologies.
- Worked on communication platforms like Slack, Microsoft Teams to share files and communicate among the team

Python Developer/ Application Development Analyst
SCube soft solutions,TX

Apr 2019 - July 2021

Responsibilities:

- Led React-based frontend development, emphasizing component-based design for modularity, and engineered Python/FAST API backend functionalities, SQL Alchemy for SQL query optimization.
- Proficient in modern relational database management systems, including MySQL, PostgreSQL.
- Implemented state management in React applications using both the Context API and hooks for modular state handling, and Redux for more complex and centralized state management, ensuring seamless data flow and integration with RESTful APIs for dynamic data updates.
- Managed simple and complex triggers within Quartz, including time-based, interval, and event-based triggers, to enhance task scheduling flexibility.
- Implemented real-time data processing pipelines using PySpark Streaming, integrating with sources like Kafka and outputting to databases or data lakes.
- Architected scalable microservices with Fast API, enabling clients to handle high traffic demands with lower latency, which significantly improved user experience and system reliability.
- Enhanced Django applications with React for a dynamic frontend, integrating Axios and Fetch API for seamless data fetching from RESTful endpoints, and utilizing Reacts state management and lifecycle methods for real-time updates, ensuring a responsive and interactive user experience.
- Optimized backend services using asynchronous programming techniques, significantly reducing server load and streamlining real-time data processing for client applications.
- Conducted rigorous unit testing with PyUnit to ensure application reliability and maintain high-quality standards.
- Configured and deployed Fastapi applications on AWS with Nginx as a reverse proxy and load balancer, integrating services like EC2 and Elastic Load Balancing for performance and scalability.
- Streamlined deployment with Docker containerization, integrating ECS for orchestration and ECR for secure Docker image storage.

Python Developer
Origin IT Solutions, Hyderabad

July 2015 – December 2017

Responsibilities:

- Utilized Flask Routes to map URL patterns to view functions, ensuring proper handling of HTTP requests.

- Developed RESTful web services in Flask, facilitating data exchange with external interfaces in JSON format.
- Implemented authentication and authorization in Flask applications using Flask-Login and Flask-HTTPAuth ensuring secure user access and session management.
- Expertise in optimizing PySpark jobs using techniques like partitioning, caching, and tuning Spark configurations for enhanced performance.
- Developed a responsive website UI with HTML, CSS, JavaScript, and AJAX, using Flask-Jinja2 for dynamic content and Flask-Bootstrap for layout organization.
- Leveraged Flask to build server-side logic and integrated DOM manipulation with JavaScript and jQuery on the client-side.
- Experienced in writing unit tests for PySpark applications using PyTest and Spark testing tools, with a focus on debugging and performance tuning.
- Implemented Flask-CORS to manage Cross-Origin Resource Sharing, facilitating secure communication between frontend and backend components.
- Implemented Flask application testing using Pytest and Flask-Testing, ensuring thorough validation of functionality and robustness.
- Designed and implemented a CI/CD pipeline using Jenkins, automating testing, building, and deployment processes for applications, reducing manual errors and deployment times while ensuring continuous delivery of new features and updates